

In case of training, the whole process is model dependent. So here is how we broadly classify machine learning algorithms:

- **Supervised machine learning algorithms**



Basic ML workflow

The algorithm analyses are known training dataset to produce an inferred function to make predictions about the output values. After sufficient training, the system can provide targets for new inputs. The machine is provided with a new set of examples so that the supervised learning algorithm analyses the training data and produces a correct outcome from the labeled data. Most machine learning models fall into

this category. It includes all the classification and regressions problems. Examples include linear regression, random forest classifier, Support Vector Machines (SVMs), Convolutional Neural Networks (CNNs).

- **Unsupervised machine learning algorithms:**

They are used when the information used to train is neither labeled nor classified. Unsupervised learning studies how systems can infer a function to describe a hidden structure from unlabeled data. Here the task of machine is to group unsorted information according to patterns, similarities and differences without any prior training data. The machine is restricted to only find hidden structures in unlabeled data and the structures do not reflect. All clustering and association algorithms fall in this category. A few common algorithms are k-means clustering, apriori algorithm for association rule mining

- **Semi-supervised machine learning algorithms**

It is a class of machine learning tasks and techniques that also make use of unlabeled data for training – typically a small amount of labeled data